

Shashank Singh

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EDUCATION

University of Lucknow

Lucknow, UP

B.Tech in Computer Science and Engineering (AI)

Oct. 2023 – Present

- **Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, Artificial Intelligence (AI), Machine Learning (ML), Cryptography.

EXPERIENCE

Industrial Training – Python & Data Science

June 2025 – July 2025

NIELIT (Govt. of India)

Remote

- Completed a 60-hour structured online training covering Python, NumPy, Pandas, Matplotlib, and introductory ML algorithms under NIELIT's curriculum.
- Built and maintained ETL/ELT data pipelines to clean, transform, and validate 76,000+ real-world property records from a large CSV dataset.
- Engineered the **House Price Predictor** capstone project (Random Forest + Streamlit) [Live Demo], actively resolving pipeline bottlenecks.

PROJECTS

DPI Engine | C++17, Network Protocols, Multithreading, PCAP

[View Output](#)

- Built a C++17 Deep Packet Inspection engine to reconstruct TCP/UDP data streams.
- Engineered a lock-free multi-threaded load-balancer and fast-path architecture for parallel processing, achieving gigabit-speed data ingestion without mutex contention.
- Demonstrated advanced problem-solving by designing efficient data handling pipelines to classify traffic via TLS SNI and HTTP Host inspection.

Your-Own-AI (VectorDB) | C++, HNSW, KD-Tree, RAG, Ollama

[View Architecture](#)

- Architected a modular C++ vector database using CMake, integrating advanced data structures (KD-Trees, HNSW).
- Designed REST APIs to integrate database data and interface seamlessly with a frontend dashboard.
- Automated the Ollama cloud GPU environment setup using Python scripts to support the local RAG pipeline and PCA frontend.

UPI-Mesh | Python, FastAPI, Cryptography, SQLAlchemy, Pytest

[Live Demo](#)

- Developed an offline-first Python/FastAPI payment backend, utilizing SQLite for data integrity and Docker for containerized deployment.
- Engineered a Zero-Trust authorization pipeline enforcing strict server-side PIN hash validation, and implemented ciphertext-hash idempotency to prevent offline replay attacks.
- Engineered hybrid RSA-2048/AES-256-GCM encryption for payments settled over a simulated Bluetooth mesh.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Libraries & Frameworks: FastAPI, SQLAlchemy, Streamlit, Pytest, Pandas, NumPy, Scikit-learn

AI & Machine Learning: Vector Databases (HNSW, KD-Tree), Retrieval-Augmented Generation (RAG), Ollama, Machine Learning Algorithms

Cybersecurity & Networking: Network Protocol Analysis (TCP/UDP/TLS), Packet Capture (PCAP), Cryptography (RSA/AES), Zero-Trust

Tools & Concepts: Docker, Git, Amazon Web Services (AWS), CI/CD, Multithreading

CERTIFICATIONS

AWS Training & Certification - Fundamentals of ML & AI

May 2026 – June 2026

Samsung Innovation Campus - Artificial Intelligence

Oct. 2025 – Dec. 2025